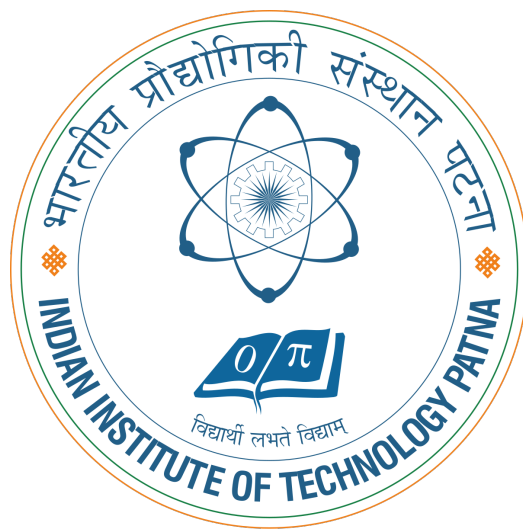


BINARY CLASSIFICATION USING KNN CLASSIFIER



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1. Objective

The objective is to build a KNN-based binary classification model from scratch for the given breast-mass dataset. The assignment requires an 80% training and 20% testing split, experimentation with $K = 3, 4, 9, 20$ and 47, and comparison of Euclidean, Manhattan, Minkowski, Cosine and Hamming distance measures. The best K and distance measure are selected using testing accuracy, followed by calculation of the confusion matrix, precision and recall. The assignment also asks for a K -versus-accuracy plot and observations from the results.

2. Dataset Description

The supplied CSV contains 569 observations and 33 columns. The target attribute is Diagnosis, where M denotes malignant and B denotes benign. Thirty numerical attributes are used as independent variables. The columns id and Unnamed: 32 are not predictive features and were removed. After preprocessing, the model uses 30 numerical features.

3. Features

radius_mean, texture_mean, perimeter_mean, area_mean, smoothness_mean, compactness_mean, concavity_mean, concave points_mean, symmetry_mean, fractal_dimension_mean, radius_se, texture_se, perimeter_se, area_se, smoothness_se, compactness_se, concavity_se, concave points_se, symmetry_se, fractal_dimension_se, radius_worst, texture_worst, perimeter_worst, area_worst, smoothness_worst, compactness_worst, concavity_worst, concave points_worst, symmetry_worst, fractal_dimension_worst.

4. Data PreProcessing

The diagnosis labels were converted to numerical form: B = 0 and M = 1. The data were randomly shuffled with a fixed random seed of 42 and divided into 455 training observations 80% and 114 testing observations 20%. Feature standardization was performed using only the training-set mean and standard deviation. The same training statistics were then applied to the test set. This prevents information from the test set from being used during preprocessing.

5. KNN Methodology

For a test observation, the KNN algorithm calculates its distance from every training observation, sorts the distances in ascending order, selects the K nearest observations, and predicts the class by majority voting. No pre-built KNN model from sklearn, PyTorch, or another machine-learning library was used. The implementation was written directly using basic Python and NumPy operations.

6. Distance Measures

- Euclidean : $d(x, y) = \sqrt{\sum (x_i - y_i)^2}$.
- Manhattan : $d(x, y) = \sum |x_i - y_i|$
- Cosine distance : $d(x, y) = 1 - \frac{\langle x, y \rangle}{(\|x\| \|y\|)}$

- Hamming distance : $d(x, y) = (\frac{1}{n}) \sum \begin{cases} 1, & \text{if } x_i \neq y_i \\ 0, & \text{if } x_i = y_i \end{cases}$
- Minkowski (for p=3) : $d(x, y) = (\sum (x_i - y_i)^3)^{\frac{1}{3}}$.

7. Experimental Setup

Parameter	Setting
Train/Test split	80%/20%
K values	3, 4, 9, 20, 47
Distance Mertics	Euclidean, Manhattan, Minkowski, Cosine, Hamming
Minkowski p	3
Random seed	42
Number of Features	30

8. Results: Testing Accuracy

Table 1: Testing Accuracy for Different Values of K and Distance Metrics

K	Euclidean	Manhattan	Minkowski	Cosine	Hamming
3	94.74%	96.49%	94.74%	96.49%	71.05%
4	95.61%	96.49%	94.74%	95.61%	73.68%
9	92.98%	94.74%	92.98%	94.74%	74.56%
20	93.86%	93.86%	93.86%	95.61%	72.81%
47	93.86%	93.86%	93.86%	95.61%	58.77%

The highest testing accuracy obtained is 96.49%, using Manhattan distance with $K = 3$. Cosine distance with $K = 3$ also produced the same accuracy, so the maximum accuracy is tied between these two configurations. The reported best configuration follows the iteration order used in the notebook.

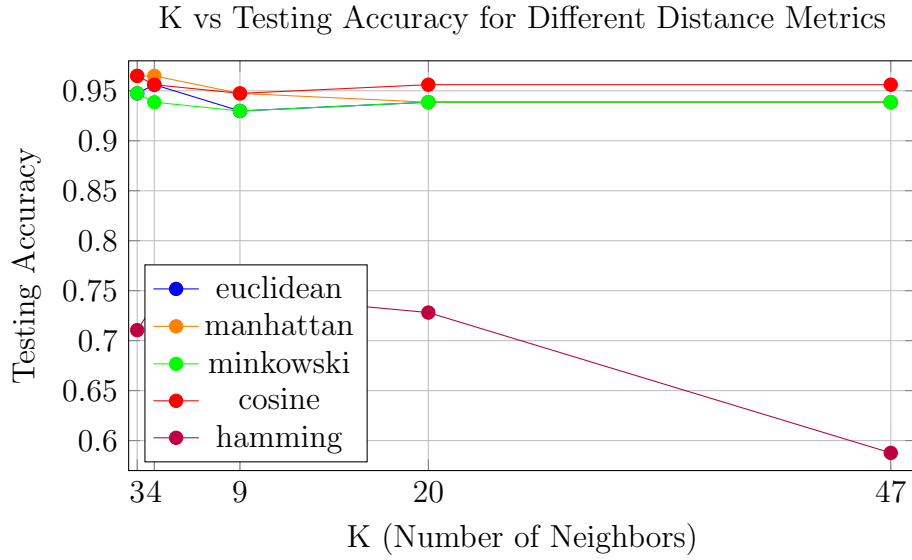


Figure 1: Testing accuracy versus K for the five distance measures.

9. Best Model Evaluation

Table 2: Confusion Matrix

	Predicted B	Predicted M
Actual B	67	0
Actual M	4	43

Accuracy: 96.49%

Precision: 100.00%

Recall: 91.49%

Specificity: 100.00%

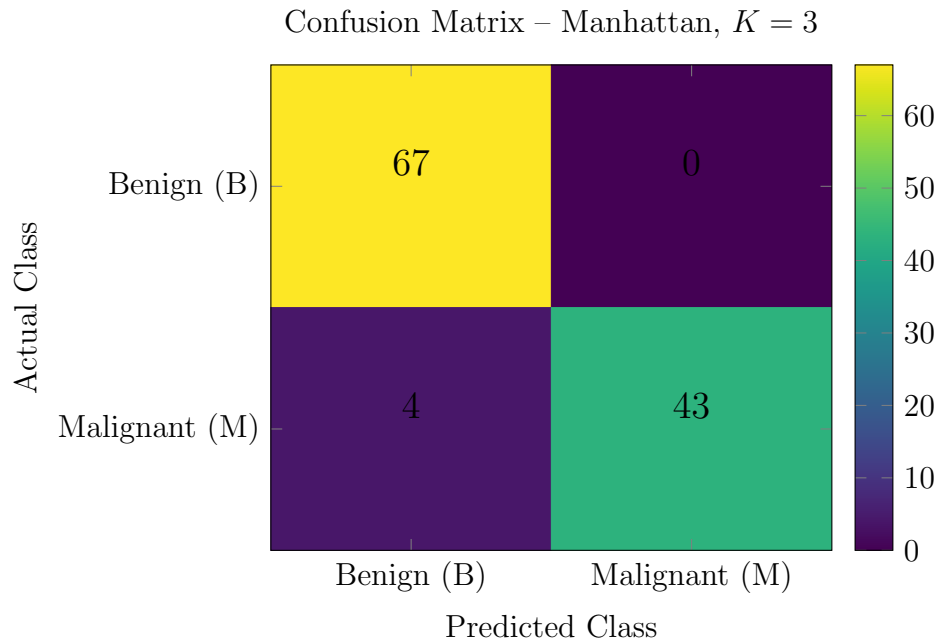


Figure 2: Confusion matrix of the selected KNN model.

10. Interpretation of Confusion Matrix

The confusion matrix contains 67 true negatives, 0 false positives, 4 false negatives and 43 true positives. Thus, all benign test observations in this particular split were classified as benign, while 4 malignant observations were classified as benign. This produces a precision of 100% for the malignant class and a recall of approximately 91.49%. The recall is lower than precision because a small number of malignant cases were missed.

11. Observations and Inferences

1. The best observed testing accuracy was 96.49% at $K = 3$ with Manhattan distance.
2. Manhattan and Cosine distances performed particularly well on this standardized continuous dataset.
3. Increasing K did not monotonically improve accuracy. For example, several metrics showed their best result at $K = 3$ or $K = 4$, while larger K values reduced or maintained accuracy.
4. A very small K can make KNN sensitive to individual observations, while a very large K can oversmooth the local neighborhood and include observations from other classes.
5. Hamming distance gave substantially lower accuracy than the other metrics in this experiment. Hamming distance is naturally designed for categorical/binary coordinate comparisons; the present features are continuous measurements. In this implementation, standardized continuous values were rounded before comparison, which limits the usefulness of Hamming distance here.
6. The plot demonstrates that the choice of distance measure can have a significant effect on KNN performance even when the same training/test split and K values are used.
7. The results are specific to the fixed random seed and 80/20 split used in this experiment. A different split can produce somewhat different test accuracies.

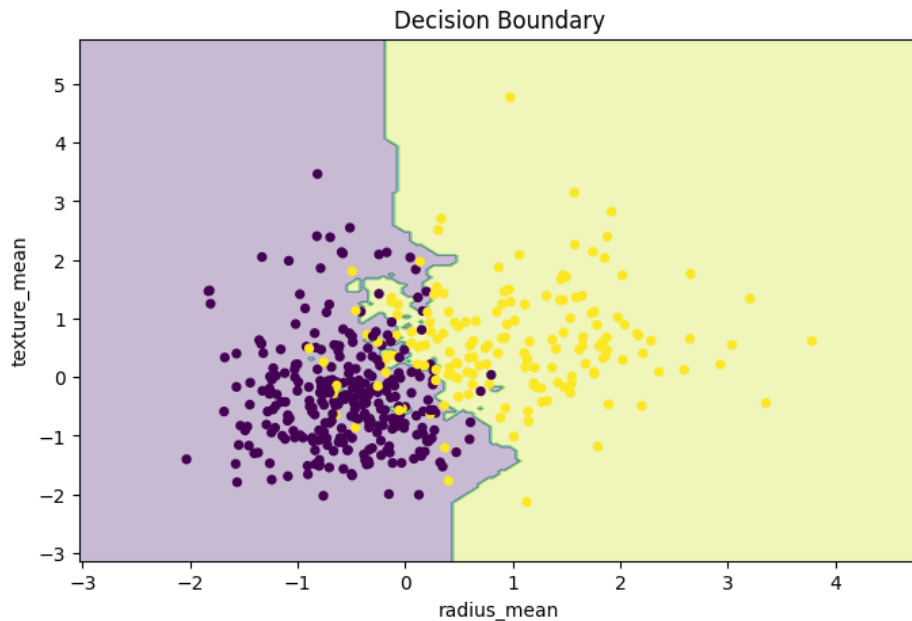


Figure 3: Decision boundary

12. Bonus: Decision Boundary Visualization

A full decision boundary cannot be directly visualized in the original 30-dimensional feature space. For the bonus visualization, the first two standardized features (Radius_mean and Texture_mean) were used to create a two-dimensional view of the KNN decision regions. This figure-3 is therefore a 2D visualization of the KNN idea and should not be interpreted as the complete 30-feature decision surface.

13. Limitations

The KNN algorithm is computationally expensive during prediction because each test observation is compared with every training observation. The present experiment also uses a single fixed 80/20 split rather than cross-

validation, because the assignment explicitly specifies the train/test division. The Hamming metric is also not naturally suited to the continuous medical measurements used here.

14. Conclusion

A binary KNN classifier was successfully implemented from scratch using the required five distance measures and five K values. For the supplied dataset and the specified random split, the highest testing accuracy was 96.49%, achieved by Manhattan distance with $K = 3$ (tied with Cosine distance at $K = 3$). The selected model achieved 100.00% precision and 91.49% recall for the malignant class. Overall, the experiment shows that both K and the distance metric have an important effect on KNN classification performance.